
Artificial Intelligence

BS (CS) Spring 2026

Lab 10 Tasks



Learning Objectives:

1. Adversarial Search
2. Alpha Beta Pruning

Lab Tasks

Submission Instructions

1. Create one notebook file for all questions with name *ROLLNO_SEC_LAB10* e.g. **i23XXXX_A_LAB10**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.

Plagiarism or heavy AI in the Lab Task will result in zero marks in the whole category.

Questions

In the game of chess, the endgame refers to the final phase where only a limited number of pieces remain on the board, and precise strategic play is required to achieve victory. Consider a simplified chess endgame scenario in which one player controls a King and a Queen, while the opponent has only a King. Implement **Alpha-Beta Pruning** in a Chess Endgame (King & Queen vs King)

In this task, you will implement an AI for a simplified chess endgame scenario where one player controls a king and queen, and the opponent has only a king. The objective is to use **Alpha-Beta pruning** to efficiently determine the best sequence of moves to checkmate the opponent in the fewest possible moves.

Your AI should:

- Use Alpha beta pruning to enhance the Minimax search and reduce unnecessary evaluations.
- Identify the best possible moves for the King and Queen to force checkmate.
- Simulate an opponent that plays optimally to survive as long as possible.

By implementing Alpha beta pruning, your AI will be able to find an optimal solution much faster than traditional Minimax.